Jackson Parrack

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Education

Saint Louis University | Saint Louis, MO — Aug, 2022 - May 2026

Bachelor of Science in Computer Engineering, Minor in Computer Science — *GPA: 3.93/4.0*

Sogang University | Seoul, South Korea | Exchange Program — Spring 2024

Relevant Coursework

Quantum Information Science and Computing, Embedded Systems, Data Structures and Algorithms, Computer Architecture, Computer Networks, Electrical and Electronic Circuits, Digital Design

Extracurricular Activities

- Tau Beta Pi Engineering Honor Society | *Member* — Apr. 2026 - May 2026
- Institute of Electrical and Electronics Engineers (IEEE) | *Vice President* — Aug. 2025 - May 2026
    - Redeveloped a dormant student chapter, increasing involvement and membership by over 200%.
    - Assisted with chapter leadership in various areas such as event organization, finance, and outreach.
- SLU Esports | *President and Teams Coordinator* — Aug. 2024 - May 2026
    - Oversaw board operations and scheduling for more than 10 club teams.
    - Increased membership to over 70 by improving leadership and engagement.

Experience

DevOps Intern | DataServ | Remote — May 2025 - Aug. 2025

- Leveraged custom Python automation scripts to migrate 3,000+ SQL stored procedures across 71 client environments into Git, allowing for easier future version control.
- Redesigned and implemented 17 post-import SQL configurations across 7 clients, replacing time-based jobs with event-driven triggers and validating reliability through full end-to-end testing.
- Optimized large-scale data ingestion processes, including a lookup redesign for 21 clients and multiple enterprise scheduling upgrades, improving system stability and performance.
- Delivered project demonstrations to leadership, communicated progress across teams, and collaborated on troubleshooting complex SQL and configuration issues in a fast-paced Agile environment.

Projects

Rapid Photonic Innovation Device (RAPID) | ECE Capstone Design Award Winner — Aug. 2025 - Apr. 2026

- Reverse-engineered laser diode assemblies, spindle and stepper motors using oscilloscopes and multimeters, and constructed motor and voice-coil driver circuits from component-level datasheets.
- Implemented motor and voice-coil control logic in VHDL on a Zynq FPGA, and built the PC-to-FPGA communication pipeline in C with a custom UART protocol and input parser.
- Built a real-time Python GUI for live pattern visualization and manual hardware control, producing a functional laser-lithography system capable of writing photonic patterns onto a spinning disc.

Vigilane | HackSLU Honorable Mention — Feb. 2026

- Developed a cross-platform mobile app using React Native (Expo) and TypeScript for live road hazard reporting.
- Implemented user submission workflows and real-time data integration between frontend and backend services.
- Built a Flask REST API with Firebase Authentication and Firestore to manage reports, sessions, and data access.

Skills

- Programming Languages: C, C++, Python, Java, JavaScript, SQL, Assembly, HTML/CSS
- Tools/Platforms: Android, iOS, AWS, GCP, Git, Linux
- Hardware: VHDL, FPGA, ARM & AVR Microcontrollers, Arduino, Raspberry Pi
- Languages: Spanish (Advanced), Korean (Intermediate), Chinese (Beginner)